

claude-windows / ce5ec24b-cf40-49bb-afb0-3a41f4cc9934

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\ce5ec24b-cf40-49bb-afb0-3a41f4cc9934\subagents\workflows\wf_191a9b86-991\agent-a70c7f688dea30812.jsonl`
- SHA-256: `aa1aa96e488ba91a1da976854c4c373eab9d9a88e4cce65cb897119c3fab7127`
- Source modified: `2026-06-22T03:49:01+00:00`
- Imported at: `2026-07-05T16:48:23+00:00`
- project: `wf_191a9b86-991`
- session_id: `ce5ec24b-cf40-49bb-afb0-3a41f4cc9934`
```

Transcript

1. user (2026-06-22T03:45:49.097Z)

Review the P2c wizard's i18n + App integration for CORRECTNESS + REGRESSION. Run `cd C:\\Users\\avidu\\Projects\\khelsutra-p2c && git diff origin/master...HEAD` and read web/src/App.tsx + the locales/*/common.json wizard additions + web/src/components/a4.touch.test.tsx in C:\\Users\\avidu\\Projects\\khelsutra-p2c. Scrutinize: (1) does adding the wizard before IngestPanel break any OTHER behavior ? the !videoId && !wizardDone vs !videoId && wizardDone gating, the videoId handoff? (2) are the wizard.* keys consistent across all 6 catalogs (en/kn/hi/es/da/id) ? same keys, same placeholders ({n}, {name})? Is the brand "KhelSutra" Latin in kn/hi (per the #77 audit)? (3) is the a4.touch.test.tsx change (skipping the wizard via the flag) legitimate, or does it weaken the test? (4) any t() key referenced by the component but missing from a catalog? Report concrete findings (file:line) only; empty list if none.

2. assistant (2026-06-22T03:45:51.415Z)

I'll start by examining the diff and the relevant files in the khelsutra-p2c repo.

3. user (2026-06-22T03:45:52.662Z)

```
locales/da/common.json      | 32 +++++
locales/en/common.json      | 32 +++++
locales/es/common.json      | 32 +++++
locales/hi/common.json      | 32 +++++
locales/id/common.json      | 32 +++++
locales/kn/common.json      | 32 +++++
web/src/App.tsx              | 9 +-
web/src/api/client.ts        | 6 ++
web/src/api/types.ts         | 17 +++
web/src/components/SetupWizard.test.tsx | 115 +++++
web/src/components/SetupWizard.tsx      | 179 +++++
web/src/components/a4.touch.test.tsx    | 4 +
12 files changed, 521 insertions(+), 1 deletion(-)
```

4. assistant (2026-06-22T03:45:54.681Z)

Let me view the full diff for the key files.

5. user (2026-06-22T03:45:56.309Z)

```
diff --git a/web/src/App.tsx b/web/src/App.tsx
index a0a7882..c516153 100644
--- a/web/src/App.tsx
+++ b/web/src/App.tsx
@@ -8,6 +8,7 @@ import { SelectionFooter } from "../components/SelectionFooter";
```

```

import { LanguageSwitcher } from "../components/LanguageSwitcher";
import { CompileResult, type BuildState } from "../components/CompileResult";
import { IngestPanel } from "../components/IngestPanel";
+import { SetupWizard, isWizardDone } from "../components/SetupWizard";
import { runQuery, type ParsedQuery } from "../lib/query";
import { floorRisk } from "../lib/floor";
import { reelFilename } from "../lib/format";
@@ -60,6 +61,8 @@ export default function App() {
  const [loadState, setLoadState] = useState<LoadState>(videoId ? { status: "loading" } : {
status: "demo" });
  const [reelName, setReelName] = useState("highlights");
  const [build, setBuild] = useState<BuildState>({ status: "idle" });
+ // P2c: the first-run wizard shows once per browser, before the ingest panel, when nothing's
loaded.
+ const [wizardDone, setWizardDone] = useState(isWizardDone);

  // Display order is separate from selection: a query can re-sort the cards while the shared
  // selection set (sel) stays the single source of truth for what lands in the reel.
@@ -234,9 +237,13 @@ export default function App() {
  <Trans i18nKey="app.intro" components={{ b: <strong />, i: <em /> }} />
</p>

+      /* P2c first-run wizard: before anything is loaded, walk a new user through AI-key /
compute /
+      videos (from /api/capabilities). Shows once per browser; then the ingest panel
takes over. */
+      {!videoId && !wizardDone && <SetupWizard onClose={() => setWizardDone(true)} />}
+
      /* Front-half ingest (B-P2): when no video is loaded, point the tool at a bucket URI /
host
      path ? process ? load. Once a video_id exists, the existing back-half takes over.
*/
-      {!videoId && (
+      {!videoId && wizardDone && (
        <IngestPanel
          onLoad={() => {
            setVideoId(id);
diff --git a/web/src/components/a4.touch.test.tsx b/web/src/components/a4.touch.test.tsx
index 1129650..4638b0f 100644
--- a/web/src/components/a4.touch.test.tsx
+++ b/web/src/components/a4.touch.test.tsx
@@ -11,6 +11,7 @@ import { render, screen, cleanup, fireEvent } from "@testing-library/react";
import i18n from "../i18n/index";
import App from "../App";
import { QueryBar } from "../QueryBar";
+import { WIZARD_DONE_KEY } from "../SetupWizard";

  // 44px floor: min-h-11 (buttons/inputs) or an explicit h-11 square (the ? dismiss).
  const TAP_44 = /\b(?:min-h-11|h-11)\b/;
@@ -18,6 +19,9 @@ const TAP_44 = /\b(?:min-h-11|h-11)\b/;
  beforeEach(async () => {
    vi.stubGlobal("fetch", vi.fn(() => Promise.reject(new Error("no engine in test"))));
    window.localStorage.clear();
+ // Skip the first-run wizard (P2c) so these tests render the app proper (IngestPanel + grid)
whose
+ // touch targets they assert ? the wizard's own ?44px targets are covered in
SetupWizard.test.tsx.
+ window.localStorage.setItem(WIZARD_DONE_KEY, "true");
    await i18n.changeLanguage("en");
  });

```

6. assistant (2026-06-22T03:45:59.163Z)

Now let me read the SetupWizard component to understand the t() keys it references and the gating/handoff logic.

7. user (2026-06-22T03:45:59.894Z)

```
1      // First-run setup wizard (PACKAGING_PLAN.md P2c). Three quick steps ? AI key / compute
/ videos ?
2      // rendered from the engine's GET /api/capabilities (an honest probe of what THIS box
can do).
3      // Localized (en/kn/hi/es/da/id). The Gemini-key step is GUIDED setup, not auto-save:
the engine has
4      // no key-write endpoint yet, so we tell the user how to add the key (offline `motion`
works with none).
5      import { useEffect, useState } from "react";
6      import { useTranslation } from "react-i18next";
7      import { getCapabilities } from "../api/client";
8      import type { EngineCapabilities } from "../api/types";
9      import { log, newCorrelationId } from "../lib/log";
10
11     /** localStorage flag ? distinct from the language key ? so the wizard shows ONCE per
browser. */
12     export const WIZARD_DONE_KEY = "kheIsutra.wizard.completed";
13
14     export function isWizardDone(): boolean {
15         try {
16             return localStorage.getItem(WIZARD_DONE_KEY) === "true";
17         } catch {
18             return false; // private-mode / no storage ? show it (harmless; user can skip)
19         }
20     }
21
22     const STEPS = 3;
23
24     function KeyStep({ caps }: { caps: EngineCapabilities | null }) {
25         const { t } = useTranslation();
26         const present = caps?.gemini_key_present === true;
27         return (
28             <div>
29                 <h3 className="font-semibold">{t("wizard.key.title")}</h3>
30                 {present ? (
31                     <p className="mt-2 font-medium text-green">{t("wizard.key.connected")}</p>
32                 ) : (
33                     <div>
34                         <p className="mt-2 text-ink/80">{t("wizard.key.missing")}</p>
35                         <p className="mt-2 text-sm text-ink/70">{t("wizard.key.how")}</p>
36                         <a
37                             href="https://aistudio.google.com/apikey"
38                             target="_blank"
39                             rel="noreferrer"
40                             className="mt-2 inline-block text-sm font-medium text-green underline"
41                         >
42                             {t("wizard.key.getKey")}
43                         </a>
44                     </div>
45                 )}
46             </div>
47         );
48     }
49
50     function ComputeStep({ caps }: { caps: EngineCapabilities | null }) {
51         const { t } = useTranslation();
52         const ffmpegOk = caps?.ffmpeg?.ffmpeg === true && caps?.ffmpeg?.ffprobe === true;
53         const gpu = caps?.gpu?.nvidia_smi_present === true;
54         const detector = caps?.default_segmenter ?? "motion";
55         return (
56             <div>
57                 <h3 className="font-semibold">{t("wizard.compute.title")}</h3>
58                 <p className="mt-2 text-ink/80">{t("wizard.compute.local")}</p>
```

```

59         <ul className="mt-3 space-y-1 text-sm text-ink/70">
60             <li>{t("wizard.compute.detector", { name: detector })}</li>
61             <li>{ffmpegOk ? t("wizard.compute.ffmpegOk") :
t("wizard.compute.ffmpegMissing")}</li>
62             <li>{gpu ? t("wizard.compute.gpu") : t("wizard.compute.noGpu")}</li>
63         </ul>
64     </div>
65 );
66 }
67
68 function VideosStep() {
69     const { t } = useTranslation();
70     return (
71         <div>
72             <h3 className="font-semibold">{t("wizard.videos.title")}</h3>
73             <p className="mt-2 text-ink/80">{t("wizard.videos.body")}</p>
74             <p className="mt-2 font-medium text-green">{t("wizard.videos.ready")}</p>
75         </div>
76     );
77 }
78
79 export function SetupWizard({ onClose }: { onClose: () => void }) {
80     const { t } = useTranslation();
81     const [step, setStep] = useState(0);
82     const [caps, setCaps] = useState<EngineCapabilities | null>(null);
83     const [loading, setLoading] = useState(true);
84     const [offline, setOffline] = useState(false);
85
86     useEffect(() => {
87         let alive = true;
88         const cid = newCorrelationId();
89         getCapabilities(cid)
90             .then((c) => {
91                 if (!alive) return;
92                 setCaps(c);
93                 setLoading(false);
94             })
95             .catch((err) => {
96                 if (!alive) return;
97                 // Engine not up yet ? don't assert false capabilities; offer a graceful skip.
98                 setOffline(true);
99                 setLoading(false);
100                 log("warn", "wizard.capabilities_unavailable", { cid, error: String(err) });
101             });
102         return () => {
103             alive = false;
104         };
105     }, []);
106
107     const finish = () => {
108         try {
109             localStorage.setItem(WIZARD_DONE_KEY, "true");
110         } catch {
111             /* private mode: still close (the wizard just reappears next visit) */
112         }
113         onClose();
114     };
115
116     const stepping = !loading && !offline; // the 3-step flow only when capabilities
loaded
117     const isLast = step === STEPS - 1;
118
119     return (
120         <section
121             role="dialog"

```

```

122     aria-label={t("wizard.title")}
123     className="mt-8 rounded-2xl border border-ink/10 bg-white p-6 shadow-sm"
124   >
125     <div className="flex items-center justify-between gap-3">
126       <h2 className="text-xl font-bold">{t("wizard.title")}</h2>
127       <button
128         type="button"
129         onClick={finish}
130         className="min-h-11 rounded-lg px-3 text-sm text-ink/60 hover:text-ink"
131       >
132         {t("wizard.skip")}
133       </button>
134     </div>
135     <p className="mt-1 text-sm text-ink/70">{t("wizard.intro")}</p>
136     {stepping && (
137       <p className="mt-1 text-xs font-medium text-ink/50">{t("wizard.step", { n: step
+ 1 })}</p>
138     )}
139
140     <div className="mt-5 min-h-[128px]" aria-live="polite">
141       {loading && <p className="text-ink/60">{t("wizard.checking")}</p>}
142       {!loading && offline && <p className="text-amber-700">{t("wizard.offline")}</p>}
143       {stepping && step === 0 && <KeyStep caps={caps} />}
144       {stepping && step === 1 && <ComputeStep caps={caps} />}
145       {stepping && step === 2 && <VideosStep />}
146     </div>
147
148     <div className="mt-6 flex items-center justify-between">
149       {stepping ? (
150         <>
151           <button
152             type="button"
153             disabled={step === 0}
154             onClick={() => setStep((s) => Math.max(0, s - 1))}
155             className="min-h-11 rounded-lg px-4 text-sm font-medium text-ink
disabled:opacity-40"
156           >
157             {t("wizard.back")}
158           </button>
159           <button
160             type="button"
161             onClick={() => (isLast ? finish() : setStep((s) => Math.min(STEPS - 1, s +
162             1)))}
163             className="min-h-11 rounded-lg bg-green px-5 text-sm font-semibold text-
white"
164           >
165             {isLast ? t("wizard.done") : t("wizard.next")}
166           </button>
167         </>
168       ) : (
169         <button
170           type="button"
171           onClick={finish}
172           className="min-h-11 rounded-lg bg-green px-5 text-sm font-semibold text-
white"
173         >
174           {t("wizard.done")}
175         </button>
176       )}
177     </div>
178   </section>
179   );
180 }

```

8. assistant (2026-06-22T03:46:02.734Z)

Now let me collect the full set of t() keys referenced. Let me look at the wizard JSON additions across all 6 catalogs.

9. user (2026-06-22T03:46:04.789Z)

```
=== en ===
diff --git a/locales/en/common.json b/locales/en/common.json
index aaaa01a..45bc8be 100644
--- a/locales/en/common.json
+++ b/locales/en/common.json
@@ -288,6 +288,38 @@
     }
   },
+  "wizard": {
+    "title": "Welcome to KhelSutra",
+    "intro": "Three quick steps and you're ready to turn a match into a highlights reel.",
+    "step": "Step {n} of 3",
+    "checking": "Checking what this device can do?",
+    "offline": "Couldn't reach KhelSutra ? start it, then reopen setup. You can skip for now.",
+    "back": "Back",
+    "next": "Next",
+    "skip": "Skip",
+    "done": "Get started",
+    "key": {
+      "title": "AI highlights (optional)",
+      "connected": "Gemini is connected ? you'll get AI-assisted highlights.",
+      "missing": "No AI key yet ? that's fine. KhelSutra works offline with the built-in motion
detector.",
+      "how": "Want sharper highlights? Get a free Gemini key, add GEMINI_API_KEY=? to the .env
file in your KhelSutra folder, then restart.",
+      "getKey": "Get a free Gemini key"
+    },
+    "compute": {
+      "title": "Where it runs",
+      "local": "Everything runs on this device ? your videos never leave it.",
+      "detector": "Detector: {name}",
+      "ffmpegOk": "Video tools (ffmpeg) ready.",
+      "ffmpegMissing": "ffmpeg not found ? install it so KhelSutra can cut and join clips.",
+      "gpu": "A GPU was detected ? the high-quality shuttle tracker can run here later.",
+      "noGpu": "No GPU needed ? the offline motion detector runs on any computer."
+    },
+    "videos": {
+      "title": "Add a video",
+      "body": "When you're ready, pick a video file or paste a cloud link and KhelSutra finds
the rallies for you.",
+      "ready": "You're all set."
+    }
+  },
+  "selection": {
+    "count": "{count, plural, one {# rally selected} other {# rallies selected}}"
+  }
}

=== kn ===
diff --git a/locales/kn/common.json b/locales/kn/common.json
index 98eb840..f6a0878 100644
--- a/locales/kn/common.json
+++ b/locales/kn/common.json
@@ -288,6 +288,38 @@
     }
   },
+  "wizard": {
+    "title": "KhelSutra???? ??????",
```

```

+   "intro": "???? ???? ????? ? ???? ????? ?????????? ?????? ??? ???? ?????????? ??????",
+   "step": "??? {n} / 3",
+   "checking": "? ???? ??? ?????????? ???? ??????????????????????",
+   "offline": "KhelSutra????? ?????????????? ? ?????? ?????????????, ???? ?????? ?????? ??????????
????????? ??????????????????",
+   "back": "?????",
+   "next": "?????",
+   "skip": "????????????",
+   "done": "????????????",
+   "key": {
+     "title": "AI ?????????? (????????)",
+     "connected": "Gemini ?????????????? ? ?????? AI-????? ????????????? ??????????",
+     "missing": "????? AI ?? ???? ? ??????????. KhelSutra ?????????????????? ?????
????????????????????? ???? ??????????",
+     "how": "????? ?????????????? ?????? ???? Gemini ?? ?????????, ?????? KhelSutra ?????????? .env
????????? GEMINI_API_KEY=? ??????, ??? ? ??????????????????",
+     "getKey": "???? Gemini ?? ??????"
+   },
+   "compute": {
+     "title": "????? ?????????????",
+     "local": "??????? ? ?????????? ????????????? ? ?????? ?????????? ?????????? ??????????????????",
+     "detector": "?????????: {name}",
+     "ffmpegOk": "??????? ???????? (ffmpeg) ??????",
+     "ffmpegMissing": "ffmpeg ?????????? ? KhelSutra ?????????????????? ?????????? ?????????? ??????
?????????",
+     "gpu": "GPU ?????????????? ? ?????? ?????????? ???? ?????????? ?????? ???? ??????????",
+     "noGpu": "GPU ?????????????? ? ?????????? ?????? ?????????? ?????????? ?????????????????? ??????????"
+   },
+   "videos": {
+     "title": "??????? ??????",
+     "body": "????????????, ?????? ???? ?????????? ???? ?????? ?????? ?????? ? KhelSutra ?????????
????????????????? ??????????????",
+     "ready": "????? ??????"
+   }
+ },
+ "selection": {
+   "count": "{count, plural, one {# ?????? ?????????????} other {# ?????????? ?????????????}}"}
+ }
=== hi ===
diff --git a/locales/hi/common.json b/locales/hi/common.json
index b114cae..282f9a8 100644
--- a/locales/hi/common.json
+++ b/locales/hi/common.json
@@ -288,6 +288,38 @@
     }
   },
   },
+   "wizard": {
+     "title": "KhelSutra ??? ???? ?????? ??",
+     "intro": "??? ???? ??? ? ??? ?? ?? ??? ?? ?????????? ??? ??? ?????? ?? ??? ?????? ??????",
+     "step": "??? {n} / 3",
+     "checking": "?? ?????? ???? ?? ???? ??, ?????? ?? ??? ???",
+     "offline": "KhelSutra ?? ???? ?????? ??? ? ??? ???? ?????, ??? ?????? ?????????? ?????? ??? ?????
????? ??????",
+     "back": "?????",
+     "next": "????",
+     "skip": "????????",
+     "done": "????? ??????",
+     "key": {
+       "title": "AI ?????????? (?????????)",
+       "connected": "Gemini ?????? ?? ? ?????? AI-????????? ???? ????????????? ??????????",
+       "missing": "???? ???? AI ?????? ???? ? ??? ??? ?????? KhelSutra ?????????? ???? ?????????? ?? ???
???? ?????? ???",
+       "how": "??????? ?????????????? ?????????? ?????? Gemini ?????? ???, ?????? KhelSutra ?????????? ?? .env
??????? ??? GEMINI_API_KEY=? ??????, ??? ?????????? ???? ??????",

```

```

+ "getKey": "?????? Gemini ????? ???"
+ },
+ "compute": {
+   "title": "?? ??? ???? ??",
+   "local": "?? ??? ??? ?????? ?? ????? ? ? ???? ?????? ???? ? ???? ????? ??????",
+   "detector": "?????????: {name}",
+   "ffmpegOk": "?????? ??? (ffmpeg) ??????",
+   "ffmpegMissing": "ffmpeg ????? ???? ? KhelSutra ?? ????? ?????? ?? ?????? ?? ??? ???
+ ?????? ??????",
+   "gpu": "GPU ??? ? ????-????????? ???? ???? ?????? ???? ???? ???? ???? ????",
+   "noGpu": "GPU ? ? ?????? ???? ? ?????? ???? ?????????? ???? ? ? ?????????? ? ???? ??"
+ },
+ "videos": {
+   "title": "?????? ??????",
+   "body": "????? ???? ??, ? ? ?????? ?????? ?????? ? ? ?????? ???? ?????? ???? ? KhelSutra ????
+ ???? ????????? ????????? ????",
+   "ready": "?? ????? ?????"
+ }
+ },
+ "selection": {
+   "count": "{count, plural, one {# ???? ?????} other {# ???????? ??????}}"
+ }
=== es ===
diff --git a/locales/es/common.json b/locales/es/common.json
index b888951..20191e2 100644
--- a/locales/es/common.json
+++ b/locales/es/common.json
@@ -288,6 +288,38 @@
   }
 }
 },
+ "wizard": {
+   "title": "Bienvenido a KhelSutra",
+   "intro": "Tres pasos rápidos y estarás listo para convertir un partido en un montaje de
+ jugadas.",
+   "step": "Paso {n} de 3",
+   "checking": "Comprobando qué puede hacer este dispositivo?",
+   "offline": "No se pudo conectar con KhelSutra ? inícialo y vuelve a abrir la configuración.
+ Puedes omitir por ahora.",
+   "back": "Atrás",
+   "next": "Siguiente",
+   "skip": "Omitir",
+   "done": "Empezar",
+   "key": {
+     "title": "Jugadas con IA (opcional)",
+     "connected": "Gemini está conectado ? tendrás jugadas asistidas por IA.",
+     "missing": "Aún no hay clave de IA ? no pasa nada. KhelSutra funciona sin conexión con el
+ detector de movimiento.",
+     "how": "¿Quieres mejores jugadas? Consigue una clave Gemini gratuita, añade
+ GEMINI_API_KEY=? al archivo .env de tu carpeta de KhelSutra y reinicia.",
+     "getKey": "Conseguir una clave Gemini gratuita"
+   },
+   "compute": {
+     "title": "Dónde se ejecuta",
+     "local": "Todo se ejecuta en este dispositivo ? tus vídeos nunca salen de él.",
+     "detector": "Detector: {name}",
+     "ffmpegOk": "Herramientas de vídeo (ffmpeg) listas.",
+     "ffmpegMissing": "No se encontró ffmpeg ? instálalo para que KhelSutra pueda cortar y
+ unir clips.",
+     "gpu": "Se detectó una GPU ? el rastreador de volante de alta calidad podrá ejecutarse
+ aquí más adelante.",
+     "noGpu": "No se necesita GPU ? el detector de movimiento sin conexión funciona en
+ cualquier ordenador."
+   },
+   "videos": {

```



```

+     "title": "Añadir un vídeo",
+     "body": "Cuando estés listo, elige un archivo de vídeo o pega un enlace en la nube y
KhelSutra encontrará las jugadas por ti.",
+     "ready": "Todo listo."
+   }
+ },
+   "selection": {
+     "count": "{count, plural, one {# jugada seleccionada} many {# jugadas seleccionadas} other
{# jugadas seleccionadas}}}"
+   }
=== da ===
diff --git a/locales/da/common.json b/locales/da/common.json
index 4a61fa2..73d865a 100644
--- a/locales/da/common.json
+++ b/locales/da/common.json
@@ -288,6 +288,38 @@
    }
  },
+  "wizard": {
+    "title": "Velkommen til KhelSutra",
+    "intro": "Tre hurtige trin, og du er klar til at lave en kamp om til en highlight-video.",
+    "step": "Trin {n} af 3",
+    "checking": "Tjekker hvad denne enhed kan?",
+    "offline": "Kunne ikke nå KhelSutra ? start det, og åbn opsætningen igen. Du kan springe
over for nu.",
+    "back": "Tilbage",
+    "next": "Næste",
+    "skip": "Spring over",
+    "done": "Kom i gang",
+    "key": {
+      "title": "AI-highlights (valgfrit)",
+      "connected": "Gemini er forbundet ? du får AI-assisterede highlights.",
+      "missing": "Ingen AI-nøgle endnu ? det er fint. KhelSutra virker offline med den
indbyggede bevægelsesdetektor.",
+      "how": "Vil du have skarpere highlights? Hent en gratis Gemini-nøgle, tilføj
GEMINI_API_KEY=? til .env-filen i din KhelSutra-mappe, og genstart.",
+      "getKey": "Hent en gratis Gemini-nøgle"
+    },
+    "compute": {
+      "title": "Hvor det kører",
+      "local": "Alt kører på denne enhed ? dine videoer forlader den aldrig.",
+      "detector": "Detektor: {name}",
+      "ffmpegOk": "Videoværktøjer (ffmpeg) er klar.",
+      "ffmpegMissing": "ffmpeg blev ikke fundet ? installer det, så KhelSutra kan klippe og
samle klip.",
+      "gpu": "En GPU blev fundet ? den høj kvalitets boldtracker kan køre her senere.",
+      "noGpu": "Ingen GPU nødvendig ? offline-bevægelsesdetektoren kører på enhver computer."
+    },
+    "videos": {
+      "title": "Tilføj en video",
+      "body": "Når du er klar, vælg en videofil eller indsæt et cloud-link, så finder KhelSutra
dueller for dig.",
+      "ready": "Du er klar."
+    }
+  },
+  "selection": {
+    "count": "{count, plural, one {# duel valgt} other {# dueller valgt}}}"
+  }
=== id ===
diff --git a/locales/id/common.json b/locales/id/common.json
index cb7ec4b..d8ebe4d 100644
--- a/locales/id/common.json
+++ b/locales/id/common.json
@@ -288,6 +288,38 @@
    }
  },
+  "wizard": {
+    "title": "Selamat datang di KhelSutra",
+    "intro": "Tiga langkah cepat, dan kamu siap untuk membuat pertandingan highlight-video.",
+    "step": "Langkah {n} dari 3",
+    "checking": "Memeriksa kemampuan perangkat ini.",
+    "offline": "Tidak dapat menghubungi KhelSutra ? mulai saja, dan buka pengaturan lagi. Kamu bisa
melompat ke bagian berikutnya.",
+    "back": "Kembali",
+    "next": "Selanjutnya",
+    "skip": "Lompat",
+    "done": "Mulai",
+    "key": {
+      "title": "AI-highlight (opsional)",
+      "connected": "Gemini terhubung ? kamu akan mendapatkan highlight yang dibantu AI.",
+      "missing": "Tidak ada kunci AI ? itu baik. KhelSutra bekerja offline dengan
detektor gerak bawaan.",
+      "how": "Mau dapat highlight yang lebih tajam? Dapatkan kunci Gemini gratis, tambahkan
GEMINI_API_KEY=? ke file .env di folder KhelSutra kamu, dan restart.",
+      "getKey": "Dapatkan kunci Gemini gratis"
+    },
+    "compute": {
+      "title": "Di mana dijalankan",
+      "local": "Semua berjalan di perangkat ini ? video kamu tidak akan meninggalkan perangkat ini.",
+      "detector": "Detektor: {name}",
+      "ffmpegOk": "Alat video (ffmpeg) sudah siap.",
+      "ffmpegMissing": "ffmpeg tidak ditemukan ? install saja, agar KhelSutra bisa memotong dan
menyambung klip.",
+      "gpu": "GPU ditemukan ? tracker kualitas tinggi bisa berjalan di sini nanti.",
+      "noGpu": "Tidak perlu GPU ? detektor gerak offline akan berjalan di komputer apa saja."
+    },
+    "videos": {
+      "title": "Tambahkan video",
+      "body": "Ketika kamu siap, pilih file video atau masukkan tautan cloud, agar KhelSutra
dapat menemukan pertandingan untukmu.",
+      "ready": "Kamu sudah siap."
+    }
+  },
+  "selection": {
+    "count": "{count, plural, one {# pertandingan terpilih} other {# pertandingan terpilih}}}"
+  }
=== id ===
diff --git a/locales/id/common.json b/locales/id/common.json
index cb7ec4b..d8ebe4d 100644
--- a/locales/id/common.json
+++ b/locales/id/common.json
@@ -288,6 +288,38 @@
    }
  },
+  "wizard": {
+    "title": "Selamat datang di KhelSutra",
+    "intro": "Tiga langkah cepat, dan kamu siap untuk membuat pertandingan highlight-video.",
+    "step": "Langkah {n} dari 3",
+    "checking": "Memeriksa kemampuan perangkat ini.",
+    "offline": "Tidak dapat menghubungi KhelSutra ? mulai saja, dan buka pengaturan lagi. Kamu bisa
melompat ke bagian berikutnya.",
+    "back": "Kembali",
+    "next": "Selanjutnya",
+    "skip": "Lompat",
+    "done": "Mulai",
+    "key": {
+      "title": "AI-highlight (opsional)",
+      "connected": "Gemini terhubung ? kamu akan mendapatkan highlight yang dibantu AI.",
+      "missing": "Tidak ada kunci AI ? itu baik. KhelSutra bekerja offline dengan
detektor gerak bawaan.",
+      "how": "Mau dapat highlight yang lebih tajam? Dapatkan kunci Gemini gratis, tambahkan
GEMINI_API_KEY=? ke file .env di folder KhelSutra kamu, dan restart.",
+      "getKey": "Dapatkan kunci Gemini gratis"
+    },
+    "compute": {
+      "title": "Di mana dijalankan",
+      "local": "Semua berjalan di perangkat ini ? video kamu tidak akan meninggalkan perangkat ini.",
+      "detector": "Detektor: {name}",
+      "ffmpegOk": "Alat video (ffmpeg) sudah siap.",
+      "ffmpegMissing": "ffmpeg tidak ditemukan ? install saja, agar KhelSutra bisa memotong dan
menyambung klip.",
+      "gpu": "GPU ditemukan ? tracker kualitas tinggi bisa berjalan di sini nanti.",
+      "noGpu": "Tidak perlu GPU ? detektor gerak offline akan berjalan di komputer apa saja."
+    },
+    "videos": {
+      "title": "Tambahkan video",
+      "body": "Ketika kamu siap, pilih file video atau masukkan tautan cloud, agar KhelSutra
dapat menemukan pertandingan untukmu.",
+      "ready": "Kamu sudah siap."
+    }
+  },
+  "selection": {
+    "count": "{count, plural, one {# pertandingan terpilih} other {# pertandingan terpilih}}}"
+  }
=== id ===

```

```

    }
  },
+ "wizard": {
+   "title": "Selamat datang di KhelSutra",
+   "intro": "Tiga langkah cepat dan Anda siap mengubah pertandingan menjadi cuplikan.",
+   "step": "Langkah {n} dari 3",
+   "checking": "Memeriksa kemampuan perangkat ini?",
+   "offline": "Tidak dapat menghubungi KhelSutra ? jalankan dulu, lalu buka kembali penyiapan.
Anda bisa lewati untuk sekarang.",
+   "back": "Kembali",
+   "next": "Berikutnya",
+   "skip": "Lewati",
+   "done": "Mulai",
+   "key": {
+     "title": "Cuplikan AI (opsional)",
+     "connected": "Gemini terhubung ? Anda akan mendapat cuplikan berbantuan AI.",
+     "missing": "Belum ada kunci AI ? tidak masalah. KhelSutra bekerja offline dengan detektor
gerak bawaan.",
+     "how": "Ingin cuplikan lebih baik? Dapatkan kunci Gemini gratis, tambahkan
GEMINI_API_KEY=? ke berkas .env di folder KhelSutra Anda, lalu mulai ulang.",
+     "getKey": "Dapatkan kunci Gemini gratis"
+   },
+   "compute": {
+     "title": "Tempat berjalan",
+     "local": "Semua berjalan di perangkat ini ? video Anda tidak pernah keluar darinya.",
+     "detector": "Detektor: {name}",
+     "ffmpegOk": "Alat video (ffmpeg) siap.",
+     "ffmpegMissing": "ffmpeg tidak ditemukan ? pasang agar KhelSutra dapat memotong dan
menggabungkan klip.",
+     "gpu": "GPU terdeteksi ? pelacak kok berkualitas tinggi dapat berjalan di sini nanti.",
+     "noGpu": "Tidak perlu GPU ? detektor gerak offline berjalan di komputer apa pun."
+   },
+   "videos": {
+     "title": "Tambahkan video",
+     "body": "Saat siap, pilih berkas video atau tempel tautan cloud dan KhelSutra akan
menemukan reli untuk Anda.",
+     "ready": "Anda siap."
+   }
+ },
+ "selection": {
+   "count": "{count, plural, other {# reli dipilih}}"
+ }

```

10. assistant (2026-06-22T03:46:10.193Z)

Now I have all the catalog data. Let me extract the exact set of t() keys referenced in the component and cross-check against each catalog programmatically. First, let me enumerate the keys referenced in SetupWizard.tsx.

The keys referenced are:

```

- `wizard.key.title`, `wizard.key.connected`, `wizard.key.missing`, `wizard.key.how`,
`wizard.key.getKey`
- `wizard.compute.title`, `wizard.compute.local`, `wizard.compute.detector` (placeholder
`{name}`), `wizard.compute.ffmpegOk`, `wizard.compute.ffmpegMissing`, `wizard.compute.gpu`,
`wizard.compute.noGpu`
- `wizard.videos.title`, `wizard.videos.body`, `wizard.videos.ready`
- `wizard.title`, `wizard.skip`, `wizard.intro`, `wizard.step` (placeholder `{n}`),
`wizard.checking`, `wizard.offline`, `wizard.back`, `wizard.done`, `wizard.next`

```

Let me verify each catalog has exactly these keys with a script.

11. user (2026-06-22T03:46:15.177Z)

EN wizard key count: 24

EN keys: ['back', 'checking', 'compute.detector', 'compute.ffmpegMissing', 'compute.ffmpegOk', 'compute.gpu', 'compute.local', 'compute.noGpu', 'compute.title', 'done', 'intro', 'key.connected', 'key.getKey', 'key.how', 'key.missing', 'key.title', 'next', 'offline', 'skip', 'step', 'title', 'videos.body', 'videos.ready', 'videos.title']

[en] key set MATCHES en (24 keys)
[kn] key set MATCHES en (24 keys)
[hi] key set MATCHES en (24 keys)
[es] key set MATCHES en (24 keys)
[da] key set MATCHES en (24 keys)
[id] key set MATCHES en (24 keys)

12. assistant (2026-06-22T03:46:17.817Z)

All 24 keys match across catalogs. Now let me verify placeholders consistency (`{n}` in step, `{name}` in compute.detector) and check that every key referenced in the component exists in the catalog. Let me also confirm no other t() keys are missing.

13. user (2026-06-22T03:46:24.276Z)

Referenced wizard keys in component: 24
All referenced keys present in all catalogs: True

[en] step placeholders={n} | detector placeholders={name}
[kn] step placeholders={n} | detector placeholders={name}
[hi] step placeholders={n} | detector placeholders={name}
[es] step placeholders={n} | detector placeholders={name}
[da] step placeholders={n} | detector placeholders={name}
[id] step placeholders={n} | detector placeholders={name}

14. assistant (2026-06-22T03:46:28.778Z)

All 24 referenced keys resolve in all catalogs and placeholders are consistent. Now let me verify the "KhelSutra" brand is Latin in kn/hi per the #77 audit. The diff shows kn `title` is `~"KhelSutra???? ??????"` and hi `title` is `~"KhelSutra ??? ?????? ??"`. Let me check all kn/hi wizard strings for any non-Latin rendering of the brand.

15. user (2026-06-22T03:46:31.439Z)

[kn] title: KhelSutra???? ??????
[kn] offline: KhelSutra????? ?????????????? ? ?????? ????????????, ??? ???? ???? ??????????
????????? ??????????????????
[kn] key.missing: ????? AI ?? ??? ? ??????????. KhelSutra ?????????????????? ????? ??????????????????
???? ??????????
[kn] key.how: ????? ????????????? ????? ??? Gemini ?? ??????, ????? KhelSutra ?????????? .env
??????? GEMINI_API_KEY=? ??????, ??? ??????????????????
[kn] compute.ffmpegMissing: ffmpeg ????????? ? KhelSutra ????????????????? ????????? ????????? ??????
?????????
[kn] videos.body: ??????????, ?????? ??? ?????????? ??? ?????? ?????? ?????? ? KhelSutra ?????????
????????????????? ??????????????

[hi] title: KhelSutra ??? ???? ?????? ??
[hi] offline: KhelSutra ?? ???? ?????? ??? ? ??? ???? ???? , ?? ???? ???? ???? ???? ???? ????
????
[hi] key.missing: ??? ?? AI ????? ???? ? ??? ?? ????? KhelSutra ????????? ???? ????????? ?? ???
??? ???? ??
[hi] key.how: ????? ????????????? ?????? ?????? Gemini ????? ???? , ????? KhelSutra ????????? ?? .env
????? ?? GEMINI_API_KEY=? ??????, ??? ?????? ???? ??????
[hi] compute.ffmpegMissing: ffmpeg ???? ???? ? KhelSutra ?? ?????? ?????? ?? ?????? ?? ??? ????
????????? ??????
[hi] videos.body: ????? ???? ?? , ?? ?????? ?????? ?????? ?? ?????? ?????? ?????? ???? ? KhelSutra
????? ??? ????????? ????????? ????

16. assistant (2026-06-22T03:46:36.372Z)

The brand "KhelSutra" is rendered in Latin script in all kn/hi wizard strings, consistent with the #77 audit. One thing to note in `kn.title`: `"KhelSutra???? ??????"` ? the Kannada dative suffix is glued directly onto the Latin brand without a space. Let me check how the existing (pre-existing) brand references in the kn catalog handle this, to determine whether this is a regression or matches the established convention.

17. user (2026-06-22T03:46:39.420Z)

```
[kn] result.shareText: ???? KhelSutra ???? ???? ???????????? ?????? ???? ?????? ? ??????????:  
khelsutra.guru  
[kn] ingest.intro: KhelSutra ???? ???? ???????? ??????, ??? ?????? ???????????? ?????????? ?  
????? ???? ???????? ??????????  
[kn] ingest.hint: ??????-???????? ???? (gs://?), ???? KhelSutra ?????????????? ??????????  
????? ?????? ??????  
[kn] ingest.reassure: ???????? ???????? ? ???? ?????? ?????????????? KhelSutra ????  
????????????????; ???? ???????? ???? ???????? ??????????  
[kn] ingest.fileHint: ? ???????? .mp4 ???? .mov ????? ? ?? ?????? ????????, ???? KhelSutra  
????????????? ??????????  
[kn] errors.networkError: KhelSutra ??????????????. ??? ??????????????  
[kn] errors.invalidUri: ???? ?????? ?????????? ?????? ? gs:// ???? ???? KhelSutra  
????????????? ?????? ?.  
[kn] site.reveal.shareText: KhelSutra ???? ? ?? ?????? ?????????????? ?????????? ?????? ?????? ???  
??? ?????????? ? khelsutra.guru  
[kn] site.hero.lead: KhelSutra ???? ???????????? ?????????? ?????????? ??????, ???-??????,  
????? ???? ???????????? ????????????????? ? ?????? ???? ?????? ??? ???? ?????????? ??????????  
????? ?????? ???? ??????????. ?????? ?????????? ???, ?????????????? ???.  
[kn] site.record.intro: ?????? ?????? ?????????????????? ??????????????. ?????????????? ?????  
??? ?????????? ???? ? ???????????? ?????? KhelSutra ?? ?????? ?????? ??????????  
[kn] site.record.upload.body: ????? ???? , ??? <b>??? ?????</b> ????? ????????? ???? ?????????  
???????? ???? ? KhelSutra ?????????????? ?????? ?????????? ???? ??????????  
[kn] site.capture.step4.b2: <b>??? ?????</b> ????? ????????? ???? ????????? ???? ?  
????????? ?????????? KhelSutra ??????????  
[kn] site.ownModel.heroLead: ????? ?????? ?????????????????? ???? Gemini ???? GPT ?????  
???????????????? ???? ??????????. KhelSutra ? ?????? ?????? ?????? ? ??? ???????????,  
???????????????? ???? ???? ????????????????? ????????? ???? ?????????? ??? ?????, <b  
style="color:#EAF3EE">????</b> ????? ???? , ???? ????????? ???? ???? ????????? ?????????.  
[kn] site.ownModel.nav.home: ? KhelSutra ??????  
[kn] site.ownModel.sample: ? <b>KhelSutra ??? ?????????????? ???? ?????????</b> ???? <a  
href="/#join">????? ????</a> ????? ???? ???? ???? ???? ??????????  
[kn] site.ownModel.train.reassureBody: KhelSutra ???? ?????-????? ?????????? ?????????? ????  
?????????, ????????? ???? ?????? ???? ???? ????????????? <b>????????</b> ?????????? ? ?????????  
?? ???? . ?????? ?????????? ???? ?????? (????) ???? <b>????????</b> ??????????: ?????  
?????????????, ???? ?????????, ???? ?????????? ?????????? ??????  
---  
[hi] result.shareText: ????? KhelSutra ?? ???? ?????????? ???????? ?? ???? ? ??????:  
khelsutra.guru  
[hi] ingest.intro: KhelSutra ?? ??? ?????? ?????? ?? ?? ???? ???? ?????? ?????? ???? ? ????? ??  
???? ? ? ???? ????  
[hi] ingest.hint: ?? ??????-???????? ???? (gs://?), ?? KhelSutra ??? ??? ???? ? ? ????? ? ? ?  
???? ??????  
[hi] ingest.reassure: ???????? ???? ?????????? ?? ? ???? ?????? ???? ???? ? ? KhelSutra ??? ????  
???? ?; ???? ?????? ? ? ???? ? ? ???? ????  
[hi] ingest.fileHint: ?? ????? ? ? ? .mp4 ? .mov ????? ? ? ? ? ? ? ? , ? KhelSutra ??????  
???? ? ? ? ?  
[hi] errors.networkError: KhelSutra ?? ???? ???? ? ? ???? ? ? ? ? ? ?  
[hi] errors.invalidUri: ???? ????? ? ? ???? ???? ? ? ? gs:// ???? ? KhelSutra ? ? ? ? ?  
? ? ? ? ?  
[hi] site.reveal.shareText: KhelSutra ????? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ?  
? ? ? ? ? ? ? khelsutra.guru  
[hi] site.hero.lead: KhelSutra ???? ???? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? , ????? ? ? ?????  
? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ?  
? ? ? ? ? ? ? ? ? ? , ? ? ? ? ? ? ? ? ? ? ? ? ?  
[hi] site.record.intro: ????? ? ? ?????????? ????????? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ?
```

```

?? ??? ?? ?????? ?????? ?? ??? KhelSutra ?? ??? ?? ??? ????? ???
[hi] site.record.upload.body: ??-???? ?????, ??? <b>???? ?????</b> ?????? ????? ?? ??? ?? ???
?? ?????? ????? ? KhelSutra ?????? ?????? ?? ?? ?????? ????? ???
[hi] site.capture.step4.b2: <b>???? ?????</b> ?????? ????? ?? ????? ?? ????? ?????? ????? ? ?????
KhelSutra ????? ???
[hi] site.ownModel.heroLead: ?? ????? ?????? ?????? ?? Gemini ?? GPT ?? ?????? ????? ?? ??????
KhelSutra ?? ????? ?????? ??? ?? ? ?? ?????, <b style="color:#EAF3EE">????</b> ????? ?????? ?????
????? ????? ??????-???? ???????? ??, ?? ??? ?????????? ????? ?????? ?? ????? ?? ??? ?????? ?? ??????????
?? ??????
[hi] site.ownModel.nav.home: ? KhelSutra ???
[hi] site.ownModel.sample: ? <b>????? ?????? ?? ?? KhelSutra ????? ?????? ???</b> ????? ?? ?? <a
href="/#join">??????? ?? ?????? ????? ???</a>, ?? ????? ?????? ??? ?????? ????? ?????? ???
[hi] site.ownModel.train.reassureBody: KhelSutra ?? ?????? ?????????? ?? ????? ??, ?????? ?????
<b>????? ?????? ?????? ??</b> ????? ?????????? ?? ?????? ?????? ??? ? ??? ?? ??? ?????? ?? ?????? ??????
????? ?? ?????? ?? ?????? ????? (????) ?? <b>?????????</b> ?????-?? ?? : ????? ??????, ????? ??, ?????
??? ?? ?????? ?? ?????? ????? ??????
---
```

18. assistant (2026-06-22T03:46:46.318Z)

This confirms the established convention: in the existing kn catalog, the brand "KhelSutra" is consistently followed by a space before Kannada postpositions (e.g. `KhelSutra ?????`, `KhelSutra ??`, `KhelSutra ?`). The wizard's new `kn.title` glues it without a space: `KhelSutra???? ??????`. This is a minor inconsistency with the established kn brand-spacing convention (and arguably harms readability of the Latin brand). The hi catalog correctly uses a space (`KhelSutra ???`). Note the wizard `kn.offline` also glues: `KhelSutra????`.

Now let me examine the App.tsx gating logic and the videoId handoff carefully. Let me read the relevant section of App.tsx in context.

19. user (2026-06-22T03:46:47.240Z)

```

55     const [videoId, setVideoId] = useState<string | null>()
56     () => new URLSearchParams(window.location.search).get("video"),
57     );
58
59     const [rallies, setRallies] = useState<RallySegment[]>(videoId ? [] : DEMO_RALLIES);
60     const [order, setOrder] = useState<RallySegment[]>(videoId ? [] : DEMO_RALLIES);
61     const [loadState, setLoadState] = useState<LoadState>(videoId ? { status: "loading" }
: { status: "demo" });
62     const [reelName, setReelName] = useState("highlights");
63     const [build, setBuild] = useState<BuildState>({ status: "idle" });
64     // P2c: the first-run wizard shows once per browser, before the ingest panel, when
nothing's loaded.
65     const [wizardDone, setWizardDone] = useState(isWizardDone);
66
67     // Display order is separate from selection: a query can re-sort the cards while the
shared
68     // selection set (sel) stays the single source of truth for what lands in the reel.
69     const sel = useRallySelection(rallies, "all");
70
71     // Load real rallies for a job's video (POST /api/query). Default ALL-selected (D8).
72     useEffect(() => {
73       if (!videoId) return;
74       const cid = newCorrelationId();
75       log("info", "rallies.load_start", { cid, video_id: videoId });
76       // Show "loading" even when videoId was set dynamically by the ingest panel (initial
state was
77       // "demo" in that path) ? the banner must reflect the load, not stale demo copy.
78       setLoadState({ status: "loading" });
79       let alive = true;
80       listRallies(videoId, cid)
81         .then((rs) => {
82           if (!alive) return;
83           setRallies(rs);
```

20. assistant (2026-06-22T03:46:55.444Z)

Now I understand the gating fully. Let me reason through the logic:

- `videoId` initial state comes from URL `?video=` param. If a URL has `?video=...`, then `!videoId` is false, so neither the wizard nor IngestPanel renders ? the back-half (loading) takes over. Correct: a deep link bypasses the wizard.
- When no `videoId` and wizard not done: wizard renders, IngestPanel does NOT (`!videoId` & `wizardDone` is false). Correct ? they're mutually exclusive.
- After wizard `onClose` ? `setWizardDone(true)`: wizard disappears, IngestPanel appears (still `!videoId`). Correct handoff.
- IngestPanel's `onLoaded` sets `videoId`, so both `!videoId` clauses become false ? back-half takes over. Correct.

One subtle point: the wizard's `finish()` writes to `localStorage` AND calls `onClose()` which sets `wizardDone` state. The `App`'s `setWizardDone(true)` callback is separate from the wizard's own `localStorage.setItem`. Both happen. No double-write issue.

Let me check: does the `IngestPanel` rendering condition change for the case where `videoId` is null but `wizardDone` is initially true (returning user)? In that case wizard never shows, IngestPanel shows immediately ? same as old behavior. Good. Let me verify the `getCapabilities` client function and types are correctly added and used.

21. user (2026-06-22T03:46:56.514Z)

```
diff --git a/web/src/api/client.ts b/web/src/api/client.ts
index fa74d2f..23c8dd4 100644
--- a/web/src/api/client.ts
+++ b/web/src/api/client.ts
@@ -7,6 +7,7 @@ import type {
  QueryResponse,
  CompileResponse,
  EngineConfig,
+ EngineCapabilities,
  ProcessResponse,
  JobStatus,
  UploadInitResponse,
@@ -133,6 +134,11 @@ export async function getConfig(cid?: string): Promise<EngineConfig> {
  return request<EngineConfig>("/config", { method: "GET", cid });
}

+/** What THIS box can do ? drives the first-run wizard. GET /api/capabilities (PACKAGING_PLAN
+P2c). */
+export async function getCapabilities(cid?: string): Promise<EngineCapabilities> {
+  return request<EngineCapabilities>("/capabilities", { method: "GET", cid });
+}
+
+/** The shot-count floor /api/compile silently enforces ? the UI must warn, not hide
+[main.py:362]. */
+export function minShotCount(cfg: EngineConfig): number | undefined {
+  return cfg.stitching?.min_shot_count;
diff --git a/web/src/api/types.ts b/web/src/api/types.ts
index 4978d7d..59749ae 100644
--- a/web/src/api/types.ts
+++ b/web/src/api/types.ts
@@ -45,6 +45,23 @@ export interface EngineConfig {
  [key: string]: unknown;
}

+/** GET /api/capabilities [backend/main.py] ? an HONEST probe of what THIS box can do, rendered
+by
+ * the first-run wizard (PACKAGING_PLAN.md P2c). Presence-booleans only (never the Gemini key
+value).
```

```

+ * Every field optional/defensive: a missing field degrades the wizard gracefully. */
+export interface EngineCapabilities {
+  api_version?: number;
+  engine_version?: string;
+  segmenters?: string[];
+  default_segmenter?: string;
+  ffmpeg?: { ffmpeg?: boolean; ffprobe?: boolean; ffmpeg_path?: string; ffprobe_path?: string };
+  gpu?: { nvidia_smi_present?: boolean };
+  /** Whether a Gemini key is PRESENT (never its value) ? drives the AI-highlights step. */
+  gemini_key_present?: boolean;
+  wasb_weights_present?: boolean;
+  deployment?: { profile?: string; compute_target?: string };
+  [key: string]: unknown;
+}
+
+ /** POST /api/process ? 202 [backend/main.py:260]. The job is queued; poll GET
+ /api/jobs/{job_id}. */
+ export interface ProcessResponse {
+   job_id: string;

```

22. assistant (2026-06-22T03:47:00.730Z)

The types are defensive (all optional). The component reads `caps?.gemini\_key\_present`, `caps?.ffmpeg?.ffmpeg`/`ffprobe`, `caps?.gpu?.nvidia\_smi\_present`, `caps?.default_segmenter ?? "motion" ? all match the type shape. Good.

Now let me examine the a4.touch.test.tsx change and the new SetupWizard.test.tsx to assess whether the test change weakens coverage.

23. user (2026-06-22T03:47:01.694Z)

```

1      // @vitest-environment jsdom
2      //
3      // A4 ? non-tech-savvy mobile UX. Two guarantees a phone user depends on:
4      //   1. TOUCH TARGETS ? 44px (WCAG 2.5.5 / platform HIG) on every primary control ?
pinned via the
5      //     Tailwind min-h-11 (=2.75rem=44px) / h-11 class so a future "tighten the padding"
can't
6      //     silently shrink a tap target below the floor.
7      //   2. The "not available yet" REASON is VISIBLE text, not a hover-only title ? a touch
device never
8      //     surfaces `title=`, so the explanation must render in the DOM for a phone user to
read it.
9      import { describe, it, expect, beforeEach, afterEach, vi } from "vitest";
10     import { render, screen, cleanup, fireEvent } from "@testing-library/react";
11     import i18n from "../i18n/index";
12     import App from "../App";
13     import { QueryBar } from "../QueryBar";
14     import { WIZARD_DONE_KEY } from "../SetupWizard";
15
16     // 44px floor: min-h-11 (buttons/inputs) or an explicit h-11 square (the ? dismiss).
17     const TAP_44 = /\b(?:min-h-11|h-11)\b/;
18
19     beforeEach(async () => {
20       vi.stubGlobal("fetch", vi.fn(() => Promise.reject(new Error("no engine in test"))));
21       window.localStorage.clear();
22       // Skip the first-run wizard (P2c) so these tests render the app proper (IngestPanel +
grid) whose
23       // touch targets they assert ? the wizard's own ?44px targets are covered in
SetupWizard.test.tsx.
24       window.localStorage.setItem(WIZARD_DONE_KEY, "true");
25       await i18n.changeLanguage("en");
26     });
27

```

```

28     afterEach(() => {
29         cleanup();
30         vi.unstubAllGlobals();
31     });
32
33     describe("A4: touch targets ? 44px on primary controls", () => {
34         it("the always-visible buttons carry a ?44px tap class", () => {
35             render(<App />);
36             for (const name of [
37                 "Select all",
38                 "Clear",
39                 "Invert",
40                 "Apply", // QueryBar submit
41                 "Process video", // IngestPanel submit
42                 "English", // LanguageSwitcher
43                 "?????",
44                 "??????",
45             ]) {
46                 const btn = screen.getByRole("button", { name });
47                 expect(btn.className, `${name} must be ?44px`).toMatch(TAP_44);
48             }
49             // The Build button label includes the live selection count ("Build reel (6)").
50             expect(screen.getByRole("button", { name: /Build reel/
51             })).className).toMatch(TAP_44);
52         });
53         it("the text inputs carry a ?44px tap class", () => {
54             render(<App />);
55             expect(screen.getByRole("textbox", { name: "Filter rallies by length, count or
56             order" })).className).toMatch(TAP_44);
57             expect(screen.getByRole("textbox", { name: "Reel name"
58             })).className).toMatch(TAP_44);
59             expect(
60                 screen.getByRole("textbox", { name: "Video location ? a gs:// bucket URL or a path
61                 on the engine host" })).className,
62             ).toMatch(TAP_44);
63         });
64         it("the sample-query chips are tappable (?44px)", () => {
65             render(<App />);
66             expect(screen.getByRole("button", { name: "over 10s" })).className).toMatch(TAP_44);
67             expect(screen.getByRole("button", { name: "top 15 longest"
68             })).className).toMatch(TAP_44);
69         });
70     });
71     describe("A4: per-card shot-count confidence is visible, not hover-only", () => {
72         it("renders the confidence inline on the chip (phones never see a title=)", () => {
73             const { container } = render(<App />); // demo rallies include a High-confidence
74             card
75             expect(container.textContent).toContain("confidence: High");
76         });
77     });
78     describe("A4: the 'not available yet' reason is visible, not hover-only", () => {
79         const HELP = "Shot type, players and score aren't available yet ? filter by rally
80         length instead.";
81         it("hidden until an unsupported concept is typed, then shown as plain text", () => {
82             render(<QueryBar onApply={() => {}} />);
83             expect(screen.queryByText(HELP)).toBeNull(); // nothing typed ? no help line
84             fireEvent.change(screen.getByRole("textbox", { name: "Filter rallies by length,
85             count or order" })), {
86                 target: { value: "smashes" }, // a shot-type token ? flagged "not available yet"
87             });
88         });
89     });

```



```

85         expect(screen.getByText(HELP)).toBeTruthy(); // the reason is now a visible DOM node
86     });
87 });
88

```

24. assistant (2026-06-22T03:47:08.011Z)

This test change is legitimate and necessary, not a weakening. Before P2c, with `localStorage` cleared and `fetch` rejecting, the App would render IngestPanel (`!videoId` true). After P2c, with `localStorage` cleared, `isWizardDone()` returns false, so the App would render the SetupWizard instead of IngestPanel ? and these tests assert on "Process video" button, the video-location textbox, etc., which live in IngestPanel. Without setting `WIZARD\_DONE\_KEY`, those `getByRole` lookups would throw and the suite would break.

So setting the flag is the correct way to restore the pre-P2c rendering path these tests target. The wizard's own 744px targets are covered separately. Let me verify that SetupWizard.test.tsx actually covers the wizard's touch targets and the gating, to confirm the comment's claim.

25. user (2026-06-22T03:47:08.804Z)

```

1      // @vitest-environment jsdom
2      import { describe, it, expect, vi, beforeEach, afterEach } from "vitest";
3      import { render, screen, fireEvent, cleanup, waitFor } from "@testing-library/react";
4      import i18n from "../i18n/index";
5      import { SetupWizard, WIZARD_DONE_KEY, isWizardDone } from "../SetupWizard";
6      import type { EngineCapabilities } from "../api/types";
7
8      function stubCapabilities(caps: EngineCapabilities | "reject") {
9          vi.stubGlobal(
10              "fetch",
11              vi.fn(async (url: string | URL) => {
12                  const u = typeof url === "string" ? url : url.toString();
13                  if (u.endsWith("/api/capabilities")) {
14                      if (caps === "reject") throw new TypeError("Failed to fetch");
15                      return { ok: true, status: 200, json: async () => caps } as unknown as Response;
16                  }
17                  return { ok: false, status: 404, text: async () => "" } as unknown as Response;
18              }),
19          );
20      }
21
22      const LIGHT: EngineCapabilities = {
23          gemini_key_present: false,
24          default_segmenter: "motion",
25          ffmpeg: { ffmpeg: false, ffprobe: false },
26          gpu: { nvidia_smi_present: false },
27      };
28      const READY: EngineCapabilities = {
29          gemini_key_present: true,
30          default_segmenter: "gemini",
31          ffmpeg: { ffmpeg: true, ffprobe: true },
32          gpu: { nvidia_smi_present: true },
33      };
34
35      beforeEach(async () => {
36          window.localStorage.clear();
37          await i18n.changeLanguage("en");
38      });
39      afterEach(() => {
40          cleanup();
41          vi.restoreAllMocks();
42          vi.unstubAllGlobals();
43      });
44
45      describe("SetupWizard", () => {

```

```

46     it("checks capabilities, then step 1 guides a missing Gemini key (offline-first
default)", async () => {
47         stubCapabilities(LIGHT);
48         render(<SetupWizard onClose={() => {}} />);
49         expect(screen.getByText(/Checking what this device can do/)).toBeTruthy();
50         expect(await screen.findByText(/built-in motion detector/)).toBeTruthy();
51         expect(screen.getByText("Get a free Gemini key")).toBeTruthy();
52         expect(screen.getByText(/Step 1 of 3/)).toBeTruthy();
53     });
54
55     it("walks all 3 steps from real capabilities; Back returns; Done sets the flag +
closes", async () => {
56         stubCapabilities(LIGHT);
57         const onClose = vi.fn();
58         render(<SetupWizard onClose={onClose} />);
59         await screen.findByText(/built-in motion detector/);
60
61         fireEvent.click(screen.getByText("Next")); // ? compute
62         expect(screen.getByText("Detector: motion")).toBeTruthy();
63         expect(screen.getByText(/ffmpeg not found/)).toBeTruthy();
64         expect(screen.getByText(/No GPU needed/)).toBeTruthy();
65
66         fireEvent.click(screen.getByText("Back")); // ? key
67         expect(screen.getByText("Get a free Gemini key")).toBeTruthy();
68
69         fireEvent.click(screen.getByText("Next")); // ? compute
70         fireEvent.click(screen.getByText("Next")); // ? videos
71         expect(screen.getByText(/You're all set/)).toBeTruthy();
72
73         fireEvent.click(screen.getByText("Get started")); // Done on last step
74         expect(onClose).toHaveBeenCalledTimes(1);
75         expect(window.localStorage.getItem(WIZARD_DONE_KEY)).toBe("true");
76     });
77
78     it("reflects a connected key + a GPU/ffmpeg-ready box", async () => {
79         stubCapabilities(READY);
80         render(<SetupWizard onClose={() => {}} />);
81         expect(await screen.findByText(/Gemini is connected/)).toBeTruthy();
82         fireEvent.click(screen.getByText("Next"));
83         expect(screen.getByText("Detector: gemini")).toBeTruthy();
84         expect(screen.getByText(/Video tools \(\ffmpeg\) ready/)).toBeTruthy();
85         expect(screen.getByText(/A GPU was detected/)).toBeTruthy();
86     });
87
88     it("Skip closes + sets the flag without finishing the steps", async () => {
89         stubCapabilities(LIGHT);
90         const onClose = vi.fn();
91         render(<SetupWizard onClose={onClose} />);
92         await screen.findByText(/built-in motion detector/);
93         fireEvent.click(screen.getByText("Skip"));
94         expect(onClose).toHaveBeenCalledTimes(1);
95         expect(isWizardDone()).toBe(true);
96     });
97
98     it("degrades gracefully when the engine is unreachable (offline) ? message + a Done
escape", async () => {
99         stubCapabilities("reject");
100        const onClose = vi.fn();
101        render(<SetupWizard onClose={onClose} />);
102        expect(await screen.findByText(/Couldn't reach KhelSutra/)).toBeTruthy();
103        // No step content / step indicator when offline.
104        expect(screen.queryByText(/Step 1 of 3/)).toBeNull();
105        fireEvent.click(screen.getByText("Get started"));
106        await waitFor(() => expect(onClose).toHaveBeenCalledTimes());
107        expect(isWizardDone()).toBe(true);

```

```

108     });
109
110     it("isWizardDone reads the localStorage flag", () => {
111         expect(isWizardDone()).toBe(false);
112         window.localStorage.setItem(WIZARD_DONE_KEY, "true");
113         expect(isWizardDone()).toBe(true);
114     });
115 });
116

```

26. assistant (2026-06-22T03:47:13.928Z)

The SetupWizard.test.tsx covers: capability fetch + step flow, Back/Next/Done navigation, connected-key path, Skip, offline degradation, and `isWizardDone`. One nuance: the comment in a4.touch.test.tsx claims "the wizard's own ?44px targets are covered in SetupWizard.test.tsx" ? but SetupWizard.test.tsx does NOT actually assert any `min-h-11`/`h-11` touch-target class. It tests behavior, not the 44px floor. So the comment slightly overstates coverage, but this is a documentation nit, not a functional defect (the wizard buttons do carry `min-h-11` in the source, lines 130, 155, 162, 171).

Now let me confirm the full test suite + lint pass to check for regression. Let me run the web tests and typecheck.

27. user (2026-06-22T03:47:15.991Z)

```

{
  "dev": "vite",
  "build": "vite build",
  "preview": "vite preview",
  "typecheck": "tsc --noEmit",
  "test": "vitest run",
  "test:coverage": "vitest run --coverage"
}

```

28. user (2026-06-22T03:47:19.014Z)

node_modules present

29. user (2026-06-22T03:47:27.886Z)

EXIT: 0

30. assistant (2026-06-22T03:47:30.585Z)

Typecheck clean. Now run the test suite, focusing on the wizard + touch tests, then the whole web suite.

31. user (2026-06-22T03:47:38.546Z)

```

{"ts":"2026-06-22T03:47:38.080Z","level":"error","event":"api.network_error","cid":"c5bf8580-1619-47eb-a957-c82c8d1e5a8a","method":"GET","url":"/api/config","error":"Error: no engine in test"}

```

stderr | src/components/a4.touch.test.tsx > A4: touch targets ? 44px on primary controls > the always-visible buttons carry a ?44px tap class

```

{"ts":"2026-06-22T03:47:38.081Z","level":"warn","event":"config.unavailable","error":"ApiError: Network error calling GET /api/config"}

```

stdout | src/components/a4.touch.test.tsx > A4: touch targets ? 44px on primary controls > the text inputs carry a ?44px tap class

```

{"ts":"2026-06-22T03:47:38.103Z","level":"info","event":"api.request","cid":"a1974c2b-2dc5-464c-acae-1f3622452e78","method":"GET","url":"/api/config"}

```

```

stderr | src/components/a4.touch.test.tsx > A4: touch targets ? 44px on
primary controls > the text inputs carry a ?44px tap class
{"ts":"2026-06-22T03:47:38.118Z","level":"error","event":"api.network_error","cid":"a1974c2b-2dc5-464c-acae-1f3622452e78","method":"GET","url":"/api/config","error":"Error: no engine in test"}

stderr | src/components/a4.touch.test.tsx > A4: touch targets ? 44px on
primary controls > the text inputs carry a ?44px tap class
{"ts":"2026-06-22T03:47:38.119Z","level":"warn","event":"config.unavailable","error":"ApiError: Network error
calling GET /api/config"}

stdout | src/components/a4.touch.test.tsx > A4: touch targets ? 44px on
primary controls > the sample-query chips are tappable (?44px)
{"ts":"2026-06-22T03:47:38.137Z","level":"info","event":"api.request","cid":"7f5cb1dc-bac7-42d3-b94a-770d2676cc16","method":"GET","url":"/api/config"}

stderr | src/components/a4.touch.test.tsx > A4: touch targets ? 44px on
primary controls > the sample-query chips are tappable (?44px)
{"ts":"2026-06-22T03:47:38.171Z","level":"error","event":"api.network_error","cid":"7f5cb1dc-bac7-42d3-b94a-770d2676cc16","method":"GET","url":"/api/config","error":"Error: no engine in
test"}

stderr | src/components/a4.touch.test.tsx > A4: touch targets ? 44px on
primary controls > the sample-query chips are tappable (?44px)
{"ts":"2026-06-22T03:47:38.171Z","level":"warn","event":"config.unavailable","error":"ApiError: Network error
calling GET /api/config"}

stdout | src/components/a4.touch.test.tsx > A4: per-card shot-count
confidence is visible, not hover-only > renders the confidence inline on the chip
(phones never see a title=)
{"ts":"2026-06-22T03:47:38.193Z","level":"info","event":"api.request","cid":"1d022b8b-6cf3-47bc-b97d-96485379c1db","method":"GET","url":"/api/config"}

stderr | src/components/a4.touch.test.tsx > A4: per-card shot-count
confidence is visible, not hover-only > renders the confidence inline on the chip
(phones never see a title=)
{"ts":"2026-06-22T03:47:38.194Z","level":"error","event":"api.network_error","cid":"1d022b8b-6cf3-47bc-b97d-96485379c1db","method":"GET","url":"/api/config","error":"Error: no engine in test"}

stderr | src/components/a4.touch.test.tsx > A4: per-card shot-count
confidence is visible, not hover-only > renders the confidence inline on the chip
(phones never see a title=)
{"ts":"2026-06-22T03:47:38.194Z","level":"warn","event":"config.unavailable","error":"ApiError: Network error
calling GET /api/config"}

? src/components/a4.touch.test.tsx (5 tests)
463ms
? A4: touch targets ? 44px on primary controls > the always-
visible buttons carry a ?44px tap class 330ms

Test Files 2 passed (2)
Tests 11 passed (11)
Start at 09:17:34
Duration 3.92s (transform 335ms, setup 36ms, collect 1.12s, tests 658ms,
environment 4.33s, prepare 520ms)

```

EXIT: 0

32. assistant (2026-06-22T03:47:40.836Z)

Both files pass (11 tests). Now run the full web suite to confirm no regression elsewhere.

33. user (2026-06-22T03:47:52.536Z)

```
{"ts":"2026-06-22T03:47:46.181Z","level":"warn","event":"upload.aborting_partial","cid":"049035d2-89b2-4239-8a0d-6abae647e3c6","upload_id":"up1","error":"ApiError: PUT /api/upload/up1/part/1 failed: 413"}
stdout | src/hooks/useIngestJob.test.ts > useIngestJob >
treats a done-but-failed pipeline verdict as emptyResult ? never a load
stdout | src/hooks/useIngestJob.test.ts > useIngestJob >
treats a done-but-failed pipeline verdict as emptyResult ? never a load
stdout | src/hooks/useIngestJob.test.ts > useIngestJob >
treats a done-but-failed pipeline verdict as emptyResult ? never a load
stdout | src/hooks/useIngestJob.test.ts > useIngestJob >
treats a done-but-failed pipeline verdict as emptyResult ? never a load
stdout | src/hooks/useIngestJob.test.ts > useIngestJob >
treats a done-but-failed pipeline verdict as emptyResult ? never a load
stderr | src/hooks/useIngestJob.test.ts > useIngestJob >
treats a done-but-failed pipeline verdict as emptyResult ? never a load
{"ts":"2026-06-22T03:47:51.700Z","level":"error","event":"ingest.job_failed","cid":"bed4e782-a4dd-4d57-80d3-6127172a7ad5","uri":"gs://b/x.mp4","code":"engine","detail":"RuntimeError: boom"}
{"ts":"2026-06-22T03:47:50.759Z","level":"warn","event":"upload.aborting_partial","cid":"5fb16d0f-3052-45b4-bd49-b28b980e4612","upload_id":"up1","error":"ApiError: PUT /api/upload/up1/part/0 failed: 413"}
{"ts":"2026-06-22T03:47:50.760Z","level":"error","event":"upload.failed","cid":"5fb16d0f-3052-45b4-bd49-b28b980e4612","filename":"match.mp4","error":"ApiError: PUT /api/upload/up1/part/0 failed: 413"}
{"ts":"2026-06-22T03:47:51.380Z","level":"warn","event":"config.unavailable","error":"ApiError: GET /api/config failed: 500"}
{"ts":"2026-06-22T03:47:51.531Z","level":"error","event":"build.failed","cid":"46b65667-4ab3-473b-88e7-463167f79c9f","video_id":"vidx","filename":"highlights.mp4","code":"networkError","error":"ApiError: Network error calling POST /api/compile"}
{"ts":"2026-06-22T03:47:51.689Z","level":"warn","event":"config.unavailable","error":"ApiError: GET /api/config failed: 500"}
{"ts":"2026-06-22T03:47:51.788Z","level":"error","event":"build.failed","cid":"ffdbc032-6c40-4b87-8c38-75d0647e2ef1","video_id":"vidx","filename":"highlights.mp4","status":400,"code":"engine","detail":"rally below min_shot_count","error":"ApiError: POST /api/compile failed: 400"}
{"ts":"2026-06-22T03:47:51.858Z","level":"error","event":"rallies.load_failed","cid":"a1d465c9-7759-4b85-89c2-6e7c4e3529d7","video_id":"vidx","code":"networkError","error":"ApiError: Network error calling POST /api/query"}
{"ts":"2026-06-22T03:47:51.859Z","level":"warn","event":"config.unavailable","error":"ApiError: GET /api/config failed: 500"}
{"ts":"2026-06-22T03:47:51.921Z","level":"warn","event":"config.unavailable","error":"ApiError: GET /api/config failed: 500"}
Test Files 24 passed (24)
Tests 221 passed (221)
EXIT: 0
```

34. assistant (2026-06-22T03:47:55.530Z)

Full web suite green: 24 files, 221 tests pass. Let me also check for any i18n key-parity test in the repo that may validate all catalogs against `en` ? to confirm there isn't a missing-key guard I should respect, and to confirm lint passes.

35. user (2026-06-22T03:47:58.315Z)

no lint script

```
---eslint config---
no eslint config in web/
```

36. user (2026-06-22T03:48:02.347Z)

```
web/src/i18n/catalog-invariants.test.ts
web/src/i18n/hygiene.test.ts
web/src/test/setup.ts
---i18n parity tests---
catalog-invariants.test.ts
hygiene.test.ts
i18n.test.tsx
index.ts
web/src/i18n/catalog-invariants.test.ts
web/src/i18n/hygiene.test.ts
```

37. assistant (2026-06-22T03:48:05.126Z)

There are i18n parity/invariant tests. They passed in the full suite (221 tests). Let me quickly confirm what `catalog-invariants.test.ts` enforces to be sure the wizard additions satisfy the key-parity and placeholder invariants the repo guards.

38. user (2026-06-22T03:48:05.593Z)

```
1      // Content regression guards for the localization catalogs ? the rules the adversarial
translation
2      // audit (#77) established, pinned so the specific bugs can never silently return. This
complements
3      // hygiene.test.ts (which proves key/placeholder PARITY) by asserting catalog CONTENT.
Each block maps
4      // to a concrete #77 bug class; the mutation that reintroduces the bug fails here. Pure
data assertions
5      // over repo-root /locales (D-CATALOG-HOME) ? no render context needed.
6      import { describe, it, expect } from "vitest";
7      import { readFileSync, readdirSync } from "node:fs";
8      import { join } from "node:path";
9
10     const LOCALES_DIR = join(import.meta.dirname, "..", "..", "..", "locales");
11
12     type Flat = Record<string, string>;
13     function flatEntries(obj: Record<string, unknown>, prefix = ""): Flat {
14         const out: Flat = {};
15         for (const [k, v] of Object.entries(obj)) {
16             const key = prefix ? `${prefix}.${k}` : k;
17             if (v && typeof v === "object" && !Array.isArray(v)) Object.assign(out,
flatEntries(v as Record<string, unknown>, key));
18             else out[key] = String(v);
19         }
20         return out;
21     }
22     const load = (loc: string): Flat =>
23         flatEntries(JSON.parse(readFileSync(join(LOCALES_DIR, loc, "common.json"), "utf8")) as
Record<string, unknown>);
24
25     const ALL_LOCALES = readdirSync(LOCALES_DIR, { withFileTypes: true })
26         .filter((d) => d.isDirectory())
27         .map((d) => d.name);
28     const en = load("en");
29
30     // (1) Internal-token / debug-tag leaks ? EVERY locale. The engine's internal
`min_shot_count` config
31     // name leaked into footer.floorSomeRest (#77, kn even wrapped it in <code>). No
developer/debug tag
32     // (kbd/samp/tt/var) or that token may reach any user-facing string. <code> is likewise
```

```

banned ?
33     // EXCEPT on the marketing site's developer "own your model" page (site.ownModel.*),
which intentionally
34     // renders inline <code> (the rally_annotator.lua filename, the ending_reason enum,
<video>.rallies.csv)
35     // through the site shim's data-i18n-html form. That namespace is the sole, deliberate
exception; the
36     // SPA console keys (where the #77 leak happened) still fail on any <code>.
37     describe("catalog content ? no internal token / debug-tag leaks (every locale)", () => {
38         const TOKEN = /min_shot_count/;
39         const DEBUG_TAG = /<\/(??:kbd|samp|tt|var)\b/i;
40         const CODE_TAG = /<\/?code\b/i;
41         const codeAllowed = (key: string) => key.startsWith("site.ownModel.");
42         for (const loc of ALL_LOCALES) {
43             it(`${loc}: no 'min_shot_count' token or stray <code>/<kbd>/<samp>/<tt>/<var> tag`,
() => {
44                 const offenders = Object.entries(load(loc))
45                     .filter(([k, v]) => TOKEN.test(v) || DEBUG_TAG.test(v) || (CODE_TAG.test(v) &&
!codeAllowed(k)))
46                     .map([k] => k);
47                 expect(offenders).toEqual([]);
48             });
49         }
50     });
51
52     // (2) Brand ? never transliterated, and present (Latin) on EVERY key en uses it on. The
positive rule
53     // is the real guard: self-maintaining (auto-covers future brand keys, incl. errors.* /
ingest.hint that
54     // the old 3-key denylist missed) and it catches ANY transliteration spelling at once ?
a transliterated
55     // value simply won't contain Latin "KheḷSutra". The Indic-spelling denylist is cheap
defence-in-depth.
56     describe("catalog content ? the brand stays Latin 'KheḷSutra' everywhere en uses it", ()
=> {
57         const brandKeys = Object.keys(en).filter((k) => en[k].includes("KheḷSutra"));
58
59         it("en actually carries the brand on several keys (so the guard is meaningful)", () =>
{
60             expect(brandKeys.length).toBeGreaterThanOrEqual(6);
61         });
62
63         for (const loc of ["kn", "hi"] as const) {
64             it(`${loc}: Latin "KheḷSutra" is present on every key whose en value uses it`, () =>
{
65                 const e = load(loc);
66                 const missing = brandKeys.filter((k) => !(e[k] ?? "").includes("KheḷSutra"));
67                 expect(missing).toEqual([]);
68             });
69         }
70
71         const TRANSLIT: Record<string, string> = { kn: "?????????", hi: "?????????" };
72         for (const loc of ["kn", "hi"] as const) {
73             it(`${loc}: the known Indic transliteration of the brand never appears`, () => {
74                 const offenders = Object.entries(load(loc))
75                     .filter([, v] => v.includes(TRANSLIT[loc]))
76                     .map([k] => k);
77                 expect(offenders).toEqual([]);
78             });
79         }
80     });
81
82     // (3) Engine/server/bucket/backend jargon only where en uses it (kn/hi). Self-
maintaining: a localized
83     // jargon term ? the Latin word OR an Indic transliteration ? is allowed ONLY on a key

```

```

whose en value
84      // also uses the English word. en avoids these in body copy (uses the brand / "cloud-
storage" / passive
85      // voice) EXCEPT the technical ingest.uriAriaLabel + uriPlaceholder; #77 leaked them
into kn/hi elsewhere.
86      describe("catalog content ? engine/server/bucket jargon only where en uses it (kn/hi)",
() => {
87          const ENGLISH = ["engine", "server", "bucket", "backend"] as const;
88          const TRANSLIT: Record<string, Record<(typeof ENGLISH)[number], string[]>> = {
89              kn: { engine: ["?????"], server: ["?????", "?????"], bucket: ["?????"], backend:
["????????"] },
90              hi: { engine: ["???"], server: ["?????", "?????"], bucket: ["?????"], backend:
["?????"] },
91          };
92          for (const loc of ["kn", "hi"] as const) {
93              it(`${loc}: a jargon term (Latin or transliterated) appears only where en uses the
English word`, () => {
94                  const e = load(loc);
95                  const violations: Array<{ key: string; term: string; en: string }> = [];
96                  for (const [key, val] of Object.entries(e)) {
97                      const enUsesWord = (w: string) => new RegExp(`\\b${w}\\b`, "i").test(en[key] ??
"");
98                      for (const w of ENGLISH) {
99                          const present = [
100                              ...(new RegExp(`\\b${w}\\b`, "i").test(val) ? [w] : []),
101                              ...TRANSLIT[loc][w].filter((t) => val.includes(t)),
102                          ];
103                          for (const term of present) if (!enUsesWord(w)) violations.push({ key, term,
en: en[key] ?? "(absent)" });
104                      }
105                  }
106                  expect(violations).toEqual([]);
107              });
108          }
109      });
110
111      // (4) Hindi de-Sanskritization (#77 class d) ? the over-Sanskritized / stiff terms must
stay replaced
112      // on the AUDITED keys. Key-scoped (not a repo-wide ban) so legitimate marketing prose
elsewhere is not
113      // constrained, while still catching a silent revert of the exact bug.
114      describe("catalog content ? Hindi de-Sanskritization stays fixed (#77 class d)", () => {
115          const hi = load("hi");
116          const BANNED: Array<[string, string]> = [
117              ["grid.shots", "????????"], // "experimental" calque
118              ["grid.shotsTitle", "????????"],
119              ["grid.shotsTitle", "????????"], // "under development" bureaucratese
120              ["query.support", "????????"],
121              ["query.support", "????????"],
122              ["grid.shotsConfidence", "????????"], // "faith" ? wrong sense for statistical
confidence
123              ["app.badge", "?????-????"], // "per-rally" calque
124          ];
125          for (const [key, term] of BANNED) {
126              it(`hi ${key} no longer contains the stiff "${term}"`, () => {
127                  expect(hi).toHaveProperty(key);
128                  expect(hi[key] ?? "").not.toContain(term);
129              });
130          }
131          it("hi keeps the friendlier replacements (??????? / ????? ?????)", () => {
132              expect(hi["grid.shots"]).toContain("????????");
133              expect(hi["grid.shotsTitle"]).toContain("????????");
134              expect(hi["query.support"]).toContain("????????");
135              expect(hi["grid.shotsConfidence"]).toContain("????? ?????");
136          });

```



```

137     });
138
139     // (5) Flagged mistranslations ? substring guard (.not.toContain) so a padded/wrapped
variant of the
140     // banned term ("????? ????????", a trailing space, ?) is still caught.
141     describe("catalog content ? flagged mistranslations stay fixed (kn/hi word choice)", ()
=> {
142         it("kn bulk.invert drops the 'rotate' mistranslation (????????)", () => {
143             expect(load("kn")["bulk.invert"]).not.toContain("????????");
144         });
145         it("kn result.dismiss drops the legalistic '????????'", () => {
146             expect(load("kn")["result.dismiss"]).not.toContain("????????");
147         });
148         it("hi result.dismiss drops the harsh '????'", () => {
149             expect(load("hi")["result.dismiss"]).not.toContain("????");
150         });
151     });
152

```

39. assistant (2026-06-22T03:48:12.820Z)

This confirms the #77 brand rule (test block 2) is exactly: Latin "KhelSutra" must be present on every key whose en value uses it, and the Indic transliteration must never appear. The wizard additions satisfy this ? every wizard key carrying the brand in en (title, offline, key.missing, key.how, compute.ffmpegMissing, videos.body) carries Latin "KhelSutra" in kn/hi, and no Indic transliteration appears. The kn `KhelSutra????` (no space) still passes this test because it `.includes("KhelSutra")``. So the no-space issue I noted is a readability nit not guarded by the suite ? confirmed minor.

Let me also confirm the hygiene.test.ts placeholder-parity guard covers the wizard's `{n}` / `{name}` (it passed, but let me verify it checks ICU/placeholder tokens so my parity claim is backed by the repo's own guard).

40. user (2026-06-22T03:48:13.702Z)

```

1      // CI guardrails that keep i18n "by design" (FRIENDLY_AND_MULTILINGUAL.md §4):
2      //   1. KEY-PARITY ? every locale catalog has EXACTLY the `en` key set (the
extensibility proof:
3      //       drop a stub locale and this lists every gap). Trivially green while only `en`
exists; it
4      //       starts enforcing the moment a kn/hi catalog lands (B5/B6).
5      //   2. NO-BARE-STRINGS (attribute level, v1) ? user-facing component attributes
(placeholder /
6      //       aria-label / title) must route through t(), never a hardcoded literal. A curated
start per
7      //       the doc; an AST-level sweep of JSX text can extend it later.
8      import { describe, it, expect } from "vitest";
9      import { readdirSync, readFileSync } from "node:fs";
10     import { join } from "node:path";
11
12     // Catalogs are hoisted to repo-root /locales (D-CATALOG-HOME): web/src/i18n ->
../.../locales.
13     const LOCALES_DIR = join(import.meta.dirname, "..", "..", "..", "locales");
14     const SRC_DIR = join(import.meta.dirname, "..");
15
16     function flatten(obj: Record<string, unknown>, prefix = ""): string[] {
17         const keys: string[] = [];
18         for (const [k, v] of Object.entries(obj)) {
19             const key = prefix ? `${prefix}.${k}` : k;
20             if (v && typeof v === "object" && !Array.isArray(v)) keys.push(...flatten(v as
Record<string, unknown>, key));
21             else keys.push(key);
22         }
23         return keys.sort();
24     }

```

```

25
26     const loadKeys = (locale: string): string[] =>
27         flatten(JSON.parse(readFileSync(join(LOCALES_DIR, locale, "common.json"), "utf8"))) as
Record<string, unknown>);
28
29     describe("i18n key-parity ? every locale has EXACTLY the en keys", () => {
30         const enKeys = loadKeys("en");
31         const others = readdirSync(LOCALES_DIR, { withFileTypes: true })
32             .filter((d) => d.isDirectory() && d.name !== "en")
33             .map((d) => d.name);
34
35         it("en is the non-empty source catalog", () => {
36             expect(enKeys.length).toBeGreaterThan(20);
37         });
38
39         for (const loc of others) {
40             it(`${loc} matches en exactly (no missing, no extra keys)`, () => {
41                 const keys = loadKeys(loc);
42                 const missing = enKeys.filter((k) => !keys.includes(k));
43                 const extra = keys.filter((k) => !enKeys.includes(k));
44                 expect({ locale: loc, missing, extra }).toEqual({ locale: loc, missing: [], extra:
[] });
45             });
46         }
47     });
48
49     // Key-parity proves every locale has the same KEYS; this proves every locale's VALUE
keeps the same
50     // simple `{placeholder}` set as en ? so a translation can never silently drop a
{count}/{error}/{min}/
51     // {filename}/{n}/? and break interpolation. (ICU plural headers like `{count, plural,
?}` carry no
52     // simple {word} placeholder and match as empty on both sides, so they're correctly
ignored.)
53     describe("placeholder parity ? every locale preserves en's {placeholders}", () => {
54         function flatEntries(obj: Record<string, unknown>, prefix = ""): Record<string,
string> {
55             const out: Record<string, string> = {};
56             for (const [k, v] of Object.entries(obj)) {
57                 const key = prefix ? `${prefix}.${k}` : k;
58                 if (v && typeof v === "object" && !Array.isArray(v)) Object.assign(out,
flatEntries(v as Record<string, unknown>, key));
59                 else out[key] = String(v);
60             }
61             return out;
62         }
63         const placeholders = (s: string) => (s.match(/\{(\w+)\}/g) ?? []).sort();
64         const load = (loc: string) => flatEntries(JSON.parse(readFileSync(join(LOCALES_DIR,
loc, "common.json"), "utf8")));
65         const enE = load("en");
66         const others = readdirSync(LOCALES_DIR, { withFileTypes: true })
67             .filter((d) => d.isDirectory() && d.name !== "en")
68             .map((d) => d.name);
69
70         for (const loc of others) {
71             it(`${loc} preserves every {placeholder} that en uses`, () => {
72                 const locE = load(loc);
73                 const mismatches: Array<{ key: string; en: string[]; got: string[] }> = [];
74                 for (const key of Object.keys(enE)) {
75                     if (!(key in locE)) continue; // missing keys are the key-parity test's job
76                     const a = placeholders(enE[key]);
77                     const b = placeholders(locE[key]);
78                     if (a.join(",") !== b.join(",")) mismatches.push({ key, en: a, got: b });
79                 }
80                 expect(mismatches).toEqual([]);

```

```

81     });
82   }
83   });
84
85   describe("es 3-form plural (CLDR one/many/other) ? MULTILINGUAL Phase A", () => {
86     // Spanish cardinal is {one, many, other}; every es ICU plural message MUST carry a
87     // (the `many` category fires on large/compact numbers) or intl-messageformat silently
88     // These are the plural-bearing keys in the catalog.
89     const es = JSON.parse(readFileSync(join(LOCALES_DIR, "es", "common.json"), "utf8")) as
Record<string, Record<string, string>>;
90     const pluralKeys: Array<[string, string]> = [
91       ["footer", "count"],
92       ["footer", "floorSomeLead"],
93       ["selection", "count"],
94     ];
95     for (const [ns, key] of pluralKeys) {
96       it(`es ${ns}.${key} carries one/many/other arms`, () => {
97         const msg = es[ns][key];
98         expect(msg, `${ns}.${key}`).toMatch(/\\bone\\s*\\{/);
99         expect(msg, `${ns}.${key}`).toMatch(/\\bmany\\s*\\{/);
100        expect(msg, `${ns}.${key}`).toMatch(/\\bother\\s*\\{/);
101      });
102    }
103  });
104
105  describe("no-bare-strings ? user-facing component attributes go through t()", () => {
106    const FILES = [
107      "App.tsx",
108      "components/QueryBar.tsx",
109      "components/TapFilters.tsx",
110      "components/RallyGrid.tsx",
111      "components/SelectionFooter.tsx",
112      "components/CompileResult.tsx",
113      "components/LanguageSwitcher.tsx",
114      "components/IngestPanel.tsx",
115    ];
116    // A hardcoded placeholder / aria-label / title with letters is a bare-string
117    // must be an expression ({t(...)}), not a quoted literal.
118    const BARE_ATTR = /\\b(?:placeholder|aria-label|title)\\s*=\\s*"\\^[^"]*"\\s*"/g;
119
120    for (const f of FILES) {
121      it(`${f}: no hardcoded placeholder/aria-label/title`, () => {
122        const src = readFileSync(join(SRC_DIR, f), "utf8");
123        expect(src.match(BARE_ATTR) ?? []).toEqual([]);
124      });
125    }
126  });
127
128  describe("Indic fonts (B4) ? self-hosted OFL Noto, no runtime CDN", () => {
129    it("main.tsx bundles the OFL Noto Kannada + Devanagari fonts (self-hosted, not Google
Fonts)", () => {
130      const main = readFileSync(join(SRC_DIR, "main.tsx"), "utf8");
131      expect(main).toMatch(/@fontsource-variable\\noto-sans-kannada/);
132      expect(main).toMatch(/@fontsource-variable\\noto-sans-devanagari/);
133    });
134    it("the font stack falls back to Noto for Kannada + Devanagari glyphs", () => {
135      const css = readFileSync(join(SRC_DIR, "index.css"), "utf8");
136      expect(css).toMatch(/Noto Sans Kannada Variable/);
137      expect(css).toMatch(/Noto Sans Devanagari Variable/);
138    });
139  });
140

```

```

141    // The Studio SPA must render correctly on an AIR-GAPPED appliance (GPU/CPU/RAM, no
internet) ?
142    // every asset (fonts, styles, scripts) is bundled local, nothing pulled from a
CDN/Google Fonts.
143    describe("offline / self-contained ? no external runtime assets", () => {
144        const CDN = /googleapis|gstatic|fonts\.google|cdnjs|unpkg|jsdelivr|bcdn\[a-z]/i;
145        const FILES = [
146            "../index.html", // web/index.html
147            "main.tsx",
148            "index.css",
149        ];
150        for (const rel of FILES) {
151            it(`${rel}: no external font/CDN reference`, () => {
152                expect(readFileSync(join(SRC_DIR, rel), "utf8")).not.toMatch(CDN);
153            });
154        }
155        it("web/index.html loads no absolute-URL <link>/<script src> (everything is
bundled/local)", () => {
156            const html = readFileSync(join(SRC_DIR, "..", "index.html"), "utf8");
157            expect(html).not.toMatch(/(?:href|src)\s*=\s*["']https?:\/\//i);
158        });
159    });
160

```

41. assistant (2026-06-22T03:48:26.732Z)

This confirms the repo's own guards back up my parity findings: key-parity and placeholder-parity both pass (and they explicitly mention `{n}` as an example placeholder).

One observation worth noting for completeness: the `no-bare-strings` guard's `FILES` list (line 105-115) does NOT include the new `components/SetupWizard.tsx`. This means a future hardcoded `placeholder`/`aria-label`/`title` literal in SetupWizard would not be caught by the guard. Currently SetupWizard.tsx uses `aria-label={t("wizard.title")}` (line 122) ? correctly routed through t() ? so there's no live bug, but the new component is outside the guard's coverage. This is a minor coverage gap, consistent with my "the comment overstates SetupWizard.test.tsx coverage" finding.

Let me do a final verification on the App.tsx logic for one edge case: does `aria-label` on the wizard dialog and the `IngestPanel` mutual-exclusion behave right when capabilities fetch fails but user is mid-wizard. That's internal to the component and tested (offline path).

I have enough to report. Let me compile the findings. All checks pass; the findings are minor/nit-level.

Summary of verification performed:

- `git diff origin/master...HEAD` reviewed for App.tsx, all 6 locales, SetupWizard.tsx/.test.tsx, a4.touch.test.tsx, client.ts, types.ts.
- Key parity: all 24 wizard keys identical across en/kn/hi/es/da/id (verified by script + repo's own hygiene.test).
- Placeholder parity: `{n}` and `{name}` consistent in all 6 (verified by script + hygiene.test).
- Brand: Latin "KheI Sutra" present in all kn/hi wizard strings; no Indic transliteration (verified + catalog-invariants.test).
- Component-referenced keys: all 24 resolve in all catalogs (no missing t() keys).
- Typecheck clean (`tsc --noEmit` exit 0); full web suite green (24 files, 221 tests).
- App.tsx gating: wizard/IngestPanel are mutually exclusive and the videoId handoff is correct.
- a4.touch.test.tsx change is a legitimate restoration of the pre-P2c render path, not a weakening.

42. user (2026-06-22T03:48:46.250Z)

Structured output provided successfully

43. assistant (2026-06-22T03:49:01.000Z)

I completed the review of the P2c wizard's i18n + App integration for correctness and regression. Here is what I verified and found.

Verdict: No blockers, no major/minor regressions. Three nits only.

What I verified (all clean)

1. ****App.tsx gating + videoId handoff**** (`web/src/App.tsx:64-65,240-256`) ? correct. The wizard (`!videoId && !wizardDone`) and IngestPanel (`!videoId && wizardDone`) are mutually exclusive; a `?video=` deep link bypasses both (back-half loads); `onClose` ? setWizardDone(true) cleanly hands off to IngestPanel; IngestPanel's `onLoaded` sets `videoId`, retiring both clauses. A returning user (`isWizardDone()` true) skips straight to IngestPanel ? same as pre-P2c. No other behavior broken.
2. ****Key parity**** ? all ****24**** `wizard.\*` keys are identical across en/kn/hi/es/da/id (scripted check + the repo's own `hygiene.test.ts` key-parity guard, both green).
3. ****Placeholder parity**** ? `{n}` (wizard.step) and `{name}` (wizard.compute.detector) present and consistent in all 6 catalogs.
4. ****Brand (#77)**** ? Latin "KhelSutra" present in every kn/hi wizard string that uses it; no Indic transliteration. Backed by `catalog-invariants.test.ts` block 2.
5. ****No missing t() keys**** ? all 24 keys referenced by SetupWizard.tsx resolve in every catalog.
6. ****Tests + types**** ? `tsc --noEmit` exit 0; full web suite green (****24 files / 221 tests****), including the wizard, touch, and i18n-invariant suites.
7. ****a4.touch.test.tsx change is legitimate**** ? setting `WIZARD\_DONE\_KEY=true` restores the pre-P2c render path (IngestPanel + grid) these tests assert on; without it the cleared-localStorage App would render the wizard and the `getByRole` lookups would throw. Not a weakening.

The three nits (returned in StructuredOutput)

- `a4.touch.test.tsx:22-23` ? comment claims the wizard's 44px targets are "covered in SetupWizard.test.tsx", but that suite tests behavior only, never the `min-h-11`/`h-11` floor. No live bug (wizard buttons carry `min-h-11`).
- `locales/kn/common.json` ? `wizard.title`/`wizard.offline` glue the suffix onto the brand without a space (`KhelSutra????`), unlike every existing kn brand reference (and unlike the correctly-spaced hi). Passes the brand guard; readability nit for native review.
- `hygiene.test.ts:105-115` ? new `SetupWizard.tsx` not added to the no-bare-strings FILES list (currently compliant, future gap).